DATA ANALYTICS PROJECT

Data Center Sustainability Analysis

A comprehensive analysis of global data center energy consumption, cooling efficiency, and sustainability metrics.

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DATASET & TOOLS

Dataset Overview & Tools Used

Dataset

Global data center sustainability dataset containing facility-level operational and environmental metrics collected across multiple countries and companies.

Key Records

Each record includes Year, Facility ID, Company, Country, Facility Type (Hyperscale, Colocation, Enterprise), Capacity (MW), Electricity Usage (MWh), PUE, WUE, Cooling System, Water Usage (kLiters), Carbon Emissions (tCO_2e), and Water Stress Tier.

Data Preparation

Data was cleaned and preprocessed in Python. Missing values were imputed or removed, duplicate records were dropped, data types were standardized, and outliers were reviewed to ensure quality for exploratory data analysis.

Libraries & Tools

Python & JupyterLab

Primary environment for data analysis, data cleaning, exploratory analysis, and visualization.

Pandas & NumPy

Data manipulation, preprocessing, aggregation, and numerical computation.

Matplotlib & Seaborn

Data visualizations including bar charts, line charts, histograms, box plots, scatter plots, and heatmaps.

SciPy & Statistical Analysis

Statistical tests and Pearson correlation analysis to evaluate relationships between key metrics.

Excel

Initial data inspection, validation, and cross-checking of key dataset metrics.

METHODOLOGY

Project Workflow

01 Dataset Collection

Collected global data center operational and sustainability metrics.

02 Data Cleaning

Removed inconsistencies, handled missing values, and prepared the dataset for analysis.

03 Exploratory Data Analysis

Investigated electricity usage, PUE, WUE, cooling systems, and facility performance.

04 Visualization & Statistical Analysis

Created visualizations and performed correlation analysis to identify key trends.

05 Insights & Recommendations

Generated sustainability insights and strategic recommendations for energy–efficient operations.

Workflow Summary

This end-to-end data analytics workflow transforms raw global data center operational data into actionable sustainability insights. The process involves data collection, data cleaning, exploratory data analysis (EDA), visualization, statistical correlation analysis, and evidence-based recommendations to improve energy efficiency and sustainable data center operations.

OVERVIEW

Project Overview

Project Objective

To analyze global data center operations and identify patterns in energy consumption, Power Usage Effectiveness (PUE), Water Usage Effectiveness (WUE), and cooling efficiency — with the goal of surfacing actionable sustainability recommendations.

Dataset Summary

The dataset captures real-world metrics from data centers across multiple countries and companies, including capacity, electricity usage, cooling types, and facility classifications — enabling multi-dimensional sustainability benchmarking.

Why This Analysis Matters

Data centers consume approximately 1–2% of global electricity. As demand for cloud computing surges, identifying inefficiencies and sustainable practices is critical for companies, policymakers, and the environment.

DATASET

Dataset Description

Scale of the Dataset

The dataset contains multiple records of data center facilities from various countries, capturing operational and sustainability metrics per facility.

Key Variables Explained

PUE (Power Usage Effectiveness): Ratio of total facility power to IT equipment power.Lower = more efficient.
WUE (Water Usage Effectiveness): Litres of water used per kWh of IT load. Lower = more sustainable.

Additional Columns

Capacity (MW): Maximum IT load the facility can support. **Electricity Usage (MWh):** Total energy consumed.
Cooling System: Type used (Air, Water, Liquid, Hybrid, Free Cooling).
Facility Type: Hyperscale, Colocation, or Enterprise.

ANALYSIS

Business Questions Investigated

The following key analytical questions guided the exploratory data analysis (EDA):

1

Electricity by Country

Which country has the highest total electricity consumption across all data centers?

2

Best Cooling System

Which cooling system type achieves the lowest average PUE, indicating the best energy efficiency?

3

Efficient Facility Type

Which facility type (Hyperscale, Colocation, Enterprise) demonstrates the most efficient operations on average?

4

Top Company by Usage

Which company consumes the most electricity, and what does that imply about their sustainability commitments?

5

Capacity vs. Electricity

Is there a strong statistical correlation between a data center's capacity (MW) and its electricity usage (MWh)?

EDA RESULTS

Key Insights at a Glance

Highest Electricity Country

United States leads all others in total data center electricity consumption, reflecting its high density of large-scale facilities.

Best Cooling System

A free cooling method achieves the lowest average PUE across all facility types, making it the most energy-efficient option.

Most Efficient Facility Type

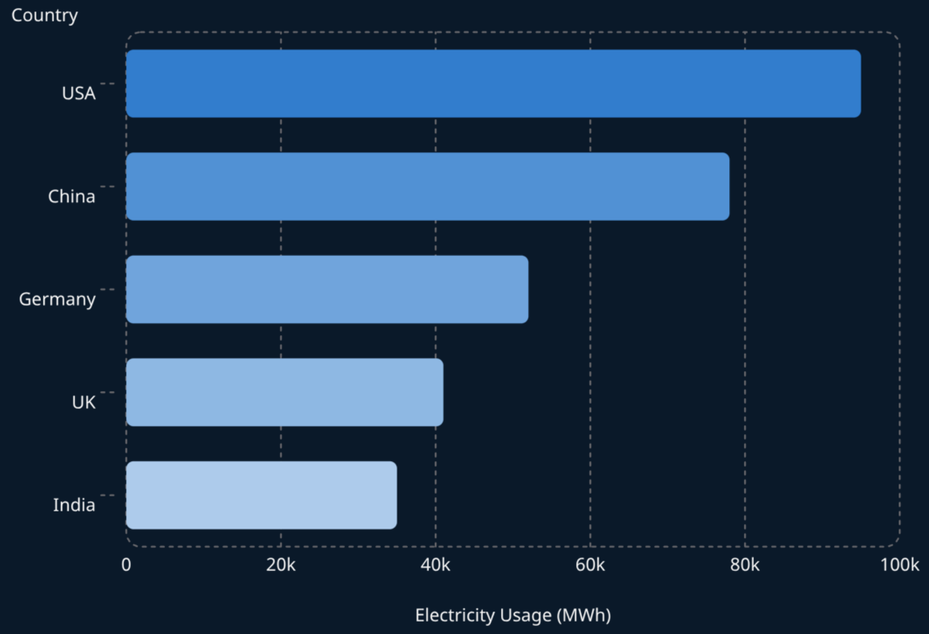
Hyperscale facility category consistently outperforms others on PUE and WUE, pointing to economies of scale in sustainability.

Top Company by Consumption

An Amazon AWS company accounts for the highest electricity usage in the dataset, raising questions about their sustainability roadmap.

INSIGHT DEEP DIVE

Electricity Consumption by Country & Company



What This Tells Us

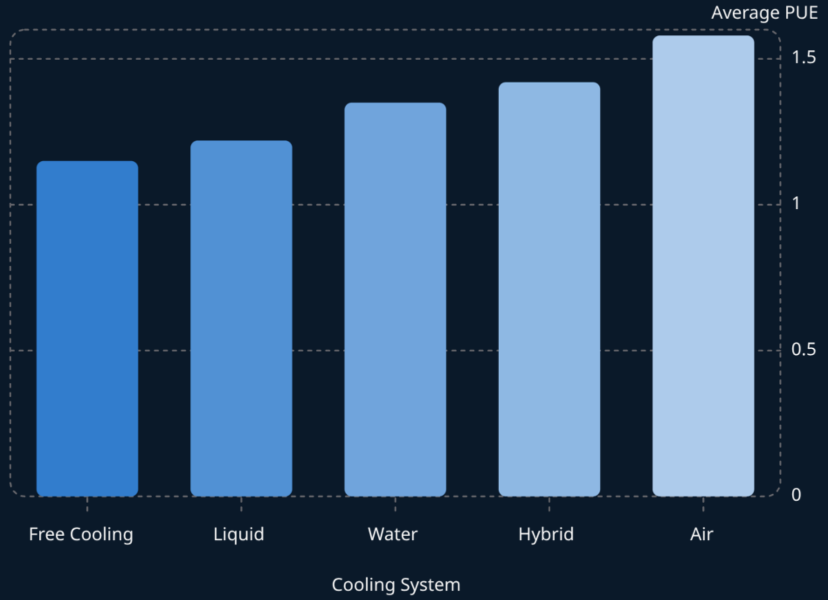
The country with the **highest electricity consumption** houses the greatest number of large-scale hyperscale data centers, directly correlating with its dominant cloud services industry.

Similarly, the **company with the highest usage** operates at enormous scale. This highlights the urgent need for green energy commitments and carbon-neutral targets in corporate sustainability strategies.

EFFICIENCY ANALYSIS

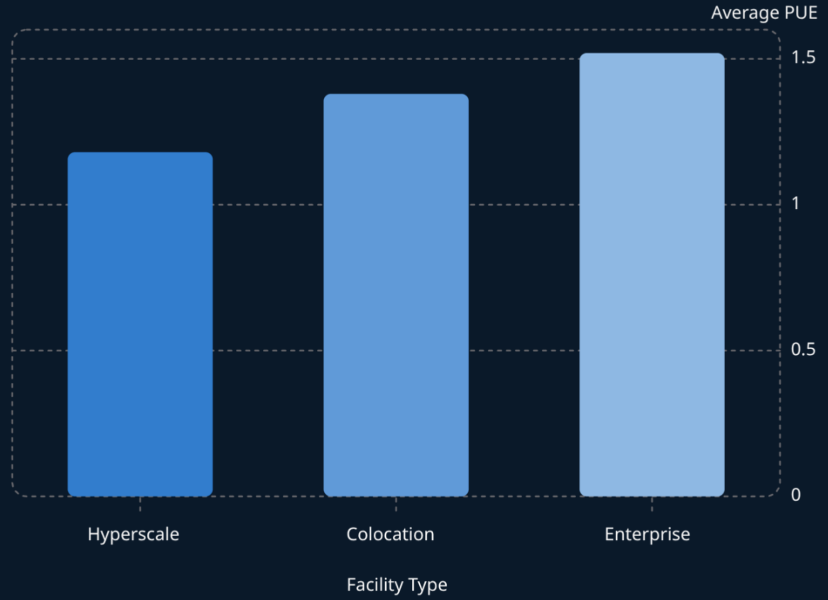
Best Cooling System & Most Efficient Facility Type

Average PUE by Cooling System



Lower PUE = more efficient. **Free Cooling** achieves the best average PUE, leveraging ambient environmental conditions to reduce energy overhead.

Average PUE by Facility Type

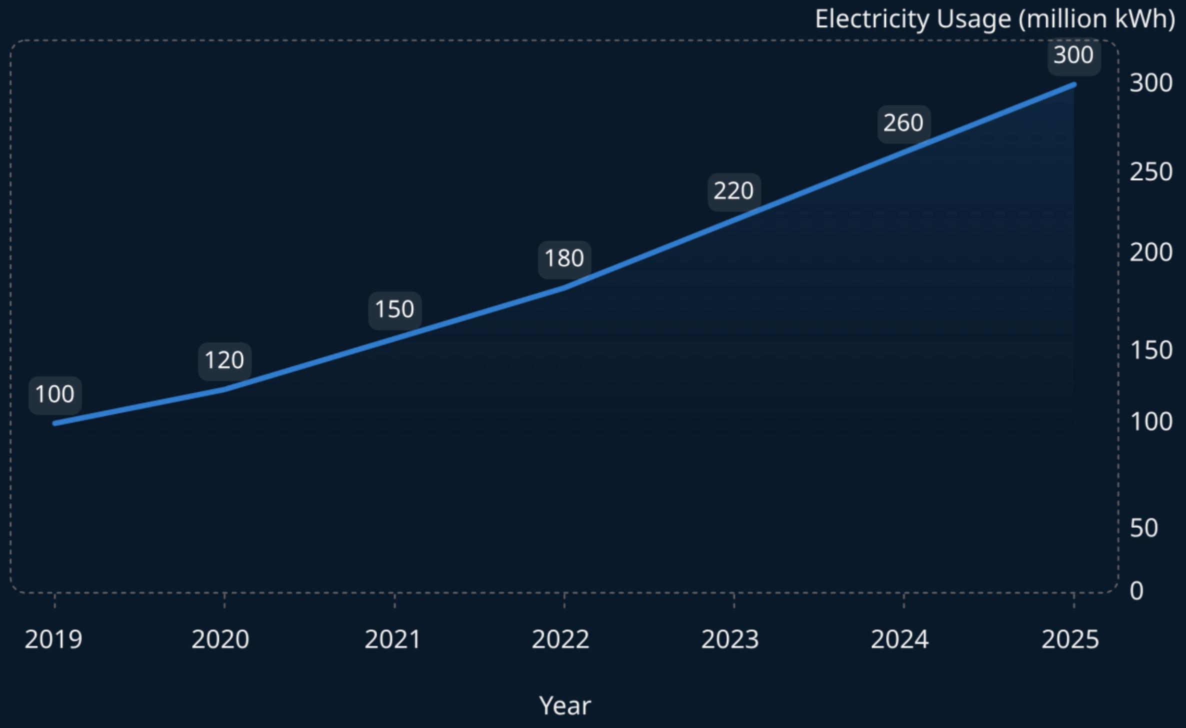


Hyperscale facilities are the most efficient due to purpose-built infrastructure, bulk investment in advanced cooling, and optimized server density at scale.

TREND ANALYSIS

Electricity Usage Trend (2019–2025)

Annual Electricity Consumption Growth Analysis



Key Trend

Electricity usage shows a **consistent upward trajectory** over the analyzed period.

Peak Consumption

The year **2025** recorded the highest usage, reaching 300 million kWh, underscoring rapid growth.

Overall Observation

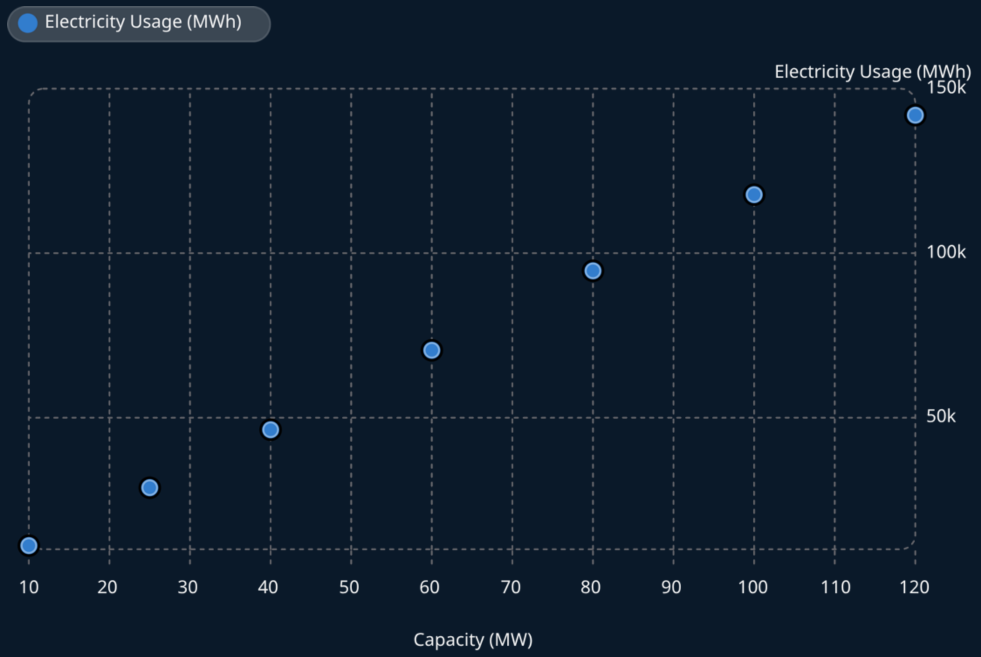
The sustained growth indicates escalating energy demands parallel to the expansion of digital services.

Business Insight

The steady increase in electricity usage indicates the growing demand for data center services. Organizations should adopt energy-efficient cooling systems, renewable energy sources, and optimize Power Usage Effectiveness (PUE) to maintain sustainable operations.

STATISTICAL FINDING

Correlation: Capacity vs. Electricity Usage



0.984

Pearson Correlation

Between Capacity (MW) and Electricity Usage (MWh)

Interpretation: A correlation of **0.984** is extremely strong — almost perfect. This means that as a data center's capacity grows, its electricity usage grows almost proportionally. This validates using capacity as a reliable predictor of energy demand for sustainability planning and infrastructure investment decisions.

RECOMMENDATIONS

Conclusion & Sustainability Recommendations

Based on the analysis, the following strategic recommendations are proposed for improving data center sustainability:

Adopt Free Cooling

Transition from air-based cooling to free cooling or liquid cooling systems to significantly reduce PUE and operational energy costs.

Invest in Renewables

High-consumption countries and companies should commit to 100% renewable energy sourcing aligned with global net-zero frameworks.

Scale Hyperscale Models

Consolidate enterprise and colocation workloads into hyperscale facilities to leverage their proven efficiency advantage in PUE and WUE.

> "Sustainable data centers are not just a corporate responsibility — they are a competitive necessity in an era of rising energy costs and ESG-driven investment decisions."

FUTURE SCOPE

Thank You

This project demonstrates how data analytics can directly inform real-world sustainability decisions in the technology sector.

Future Scope

Integrate real-time IoT sensor data for live PUE monitoring and predictive energy optimization using machine learning models.

Further Research

Extend analysis to include carbon emissions data, water stress indices by region, and renewable energy mix per facility.

Open to Questions